

AWS re:Invent

NOV. 28 – DEC. 2, 2022 | LAS VEGAS, NV

FSI314-R

Multi-Region deployments with CI/CD and Terraform on AWS

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Agenda

Architecture overview – 25 minutes

- Why multi-Region deployments?
- Pipeline architecture and Terraform code structure
- Pipeline best practices

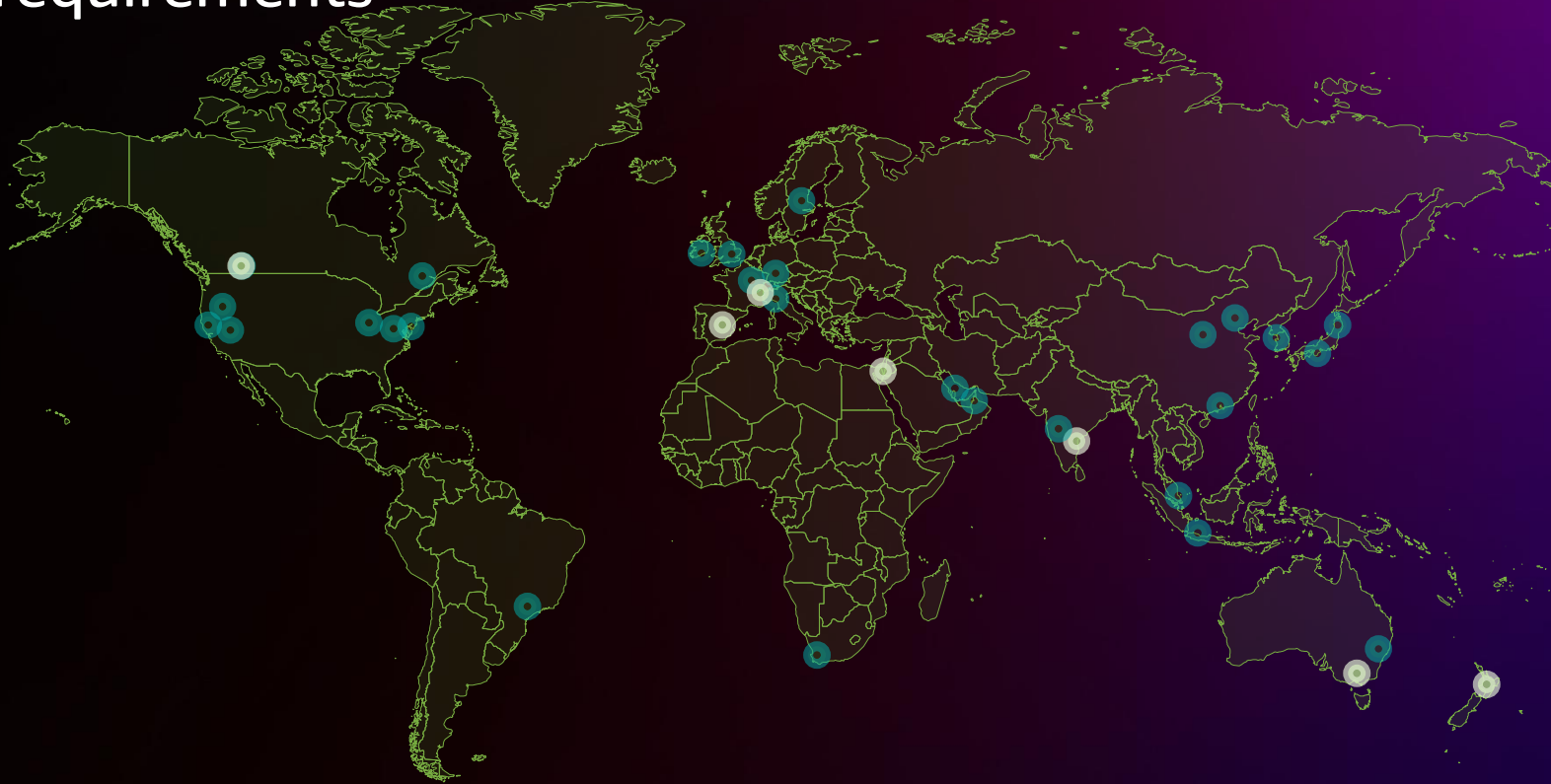
Demo – 4 minutes

Open discussion and Q&A – 30 minutes



Why multi-Region deployments?

1. Regulatory and compliance requirements
2. Mission critical applications
3. Customer experience

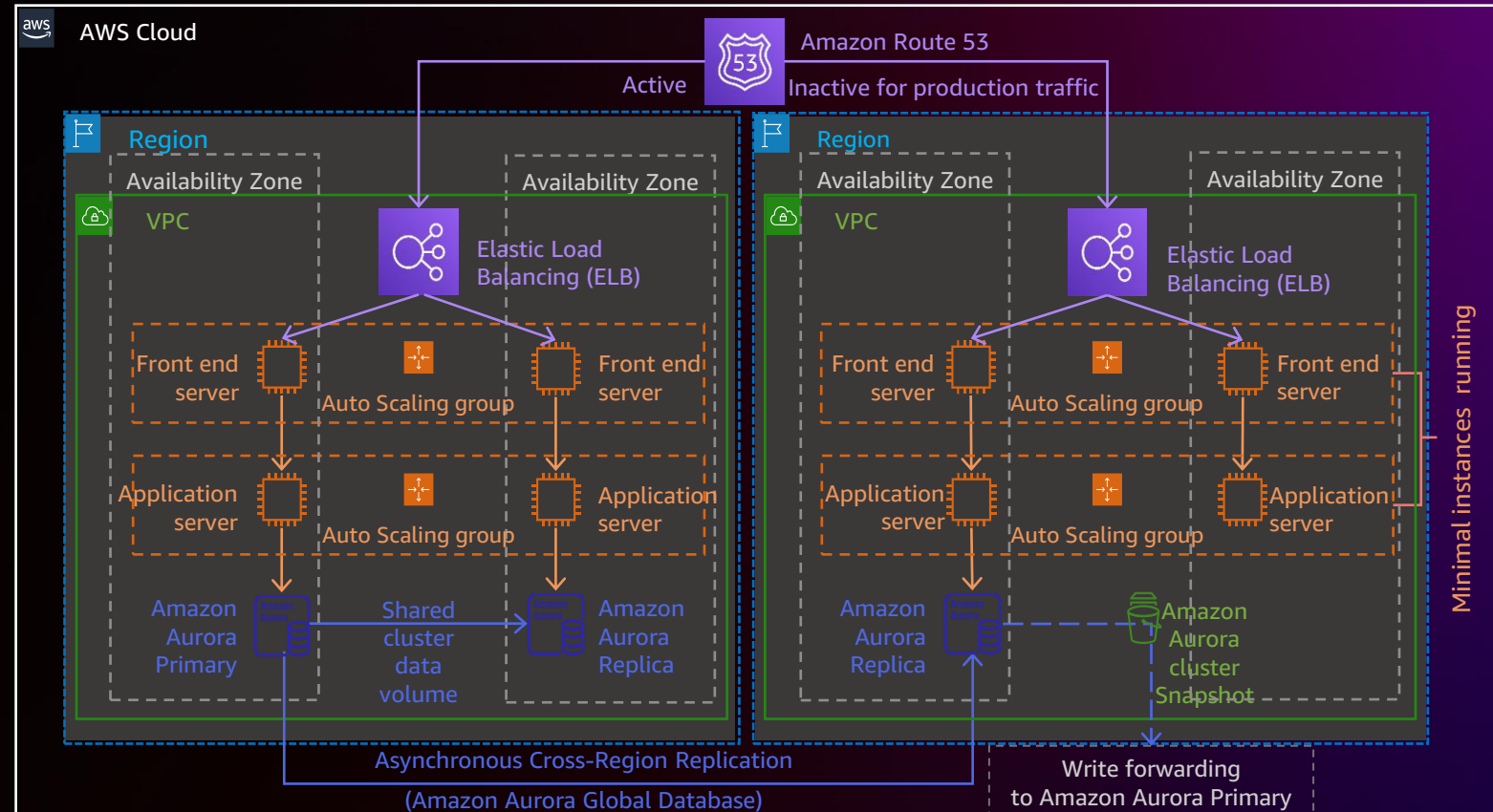


Multi-Region architectures can be complex!

Resources drift over time

This can lead to:

- Unnecessary cost
- Security incidents
- Downtime



How do we build at scale?



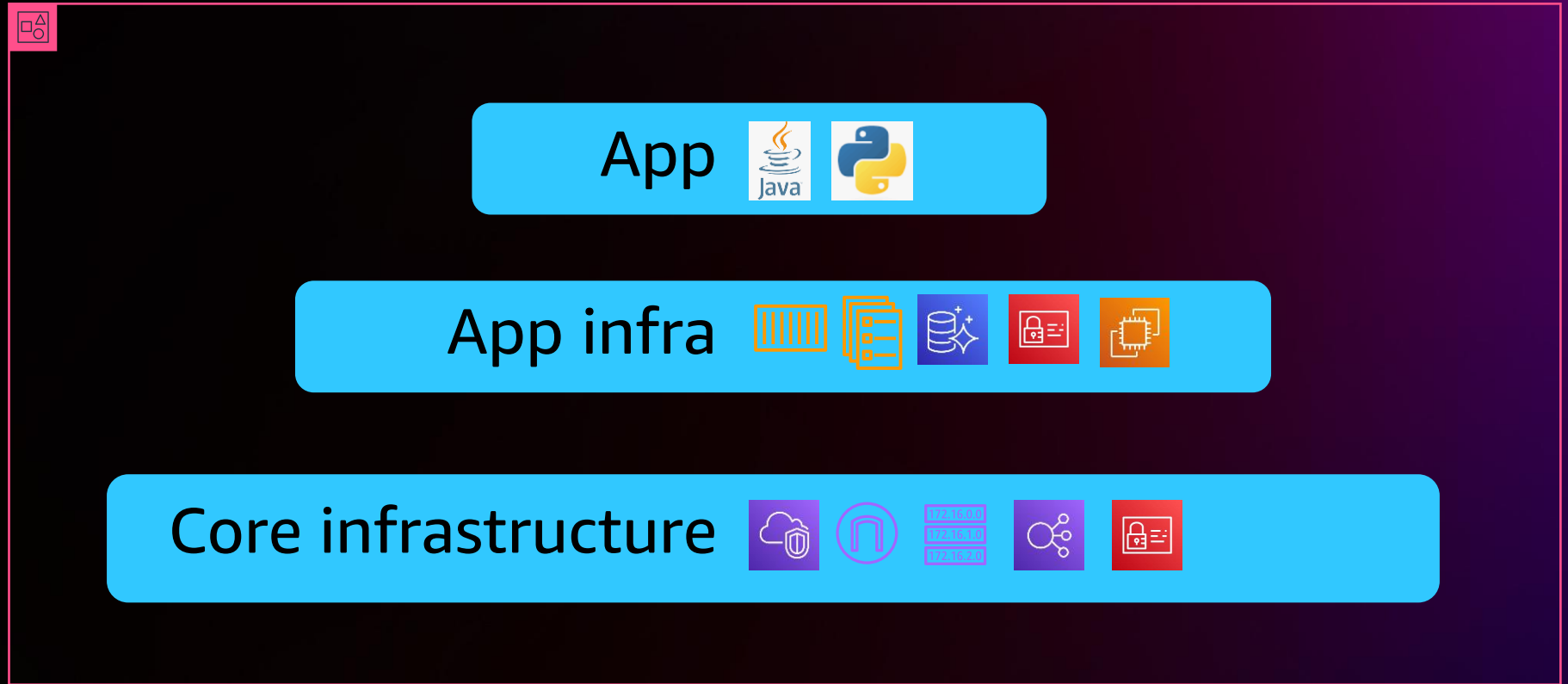
Blueprints



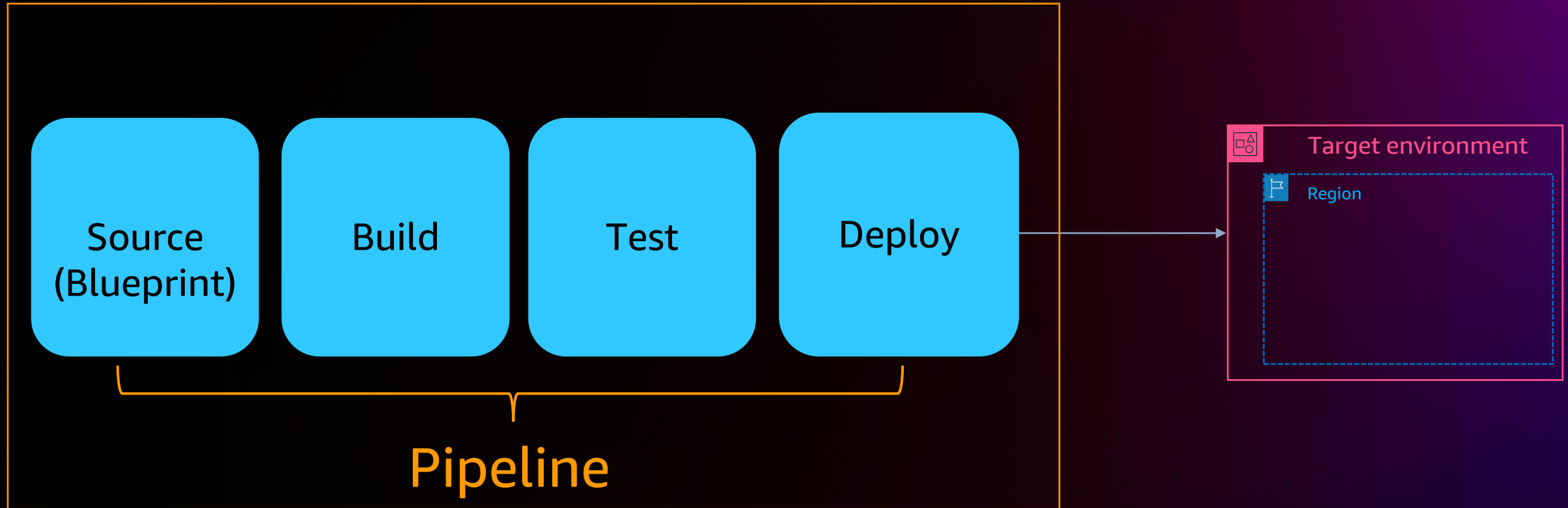
Assembly lines

Building layer by layer

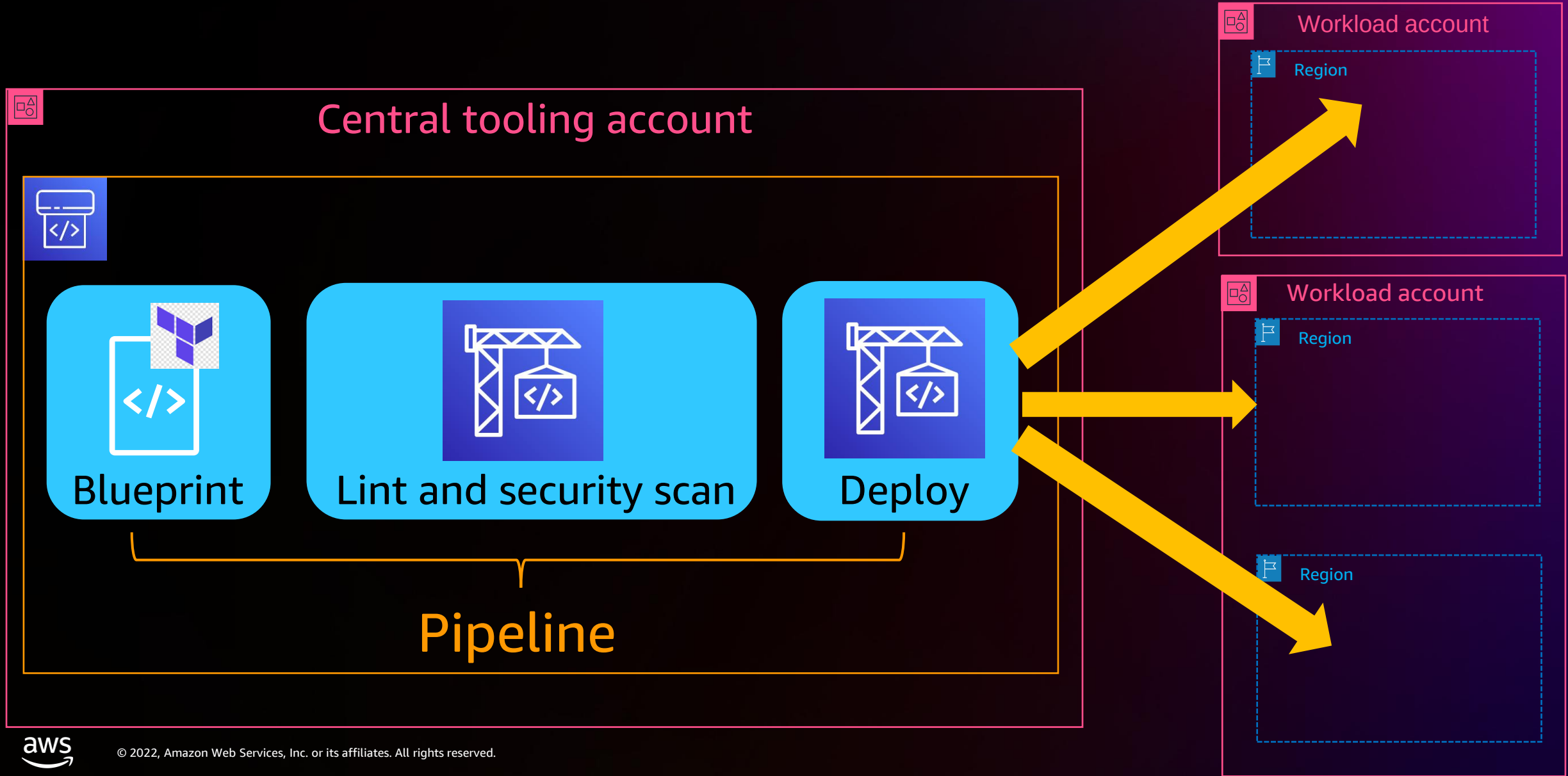
Building
bottom up



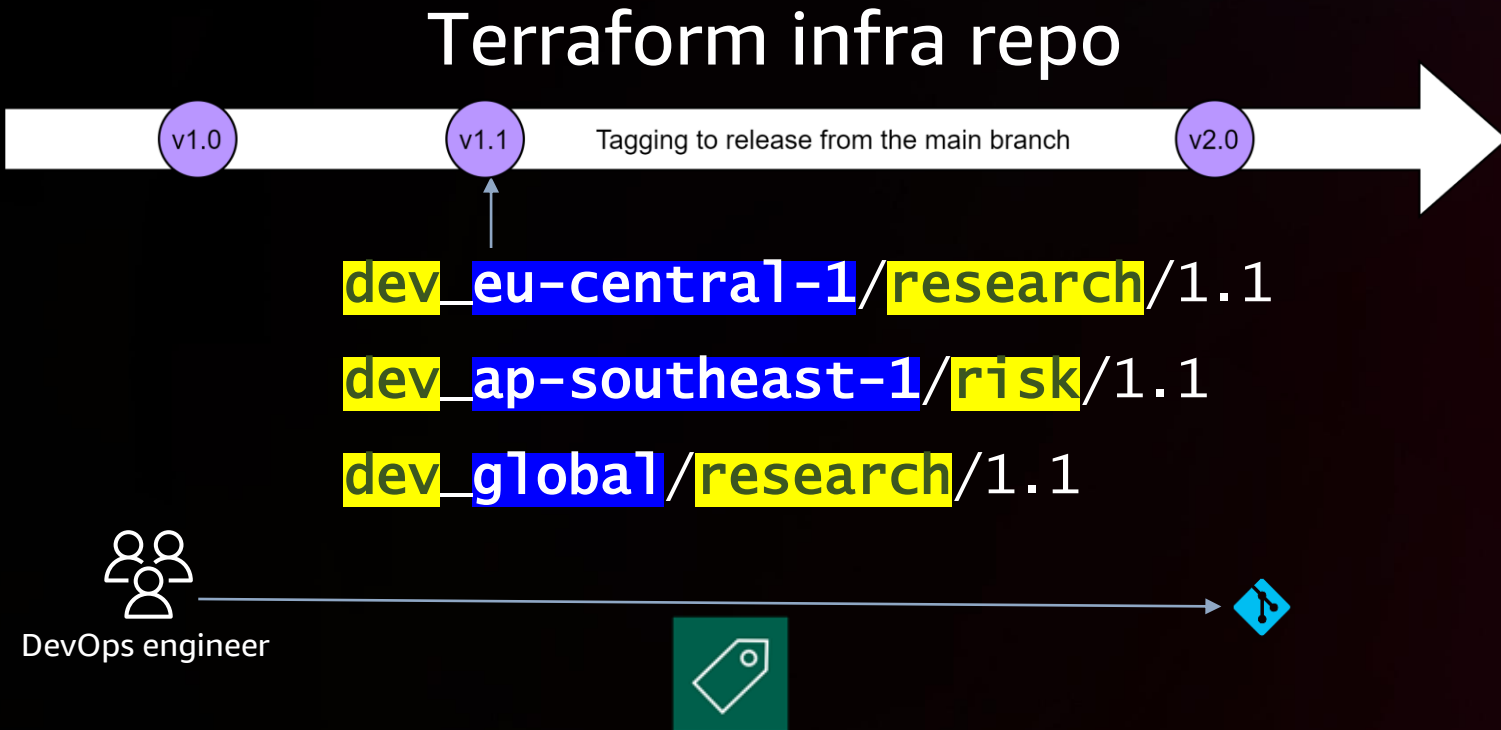
Deployment pipeline



Deployment pipeline with AWS DevOps tools



Deployments at scale



Parametrized provider

 dev eu-central-1/research/1.1

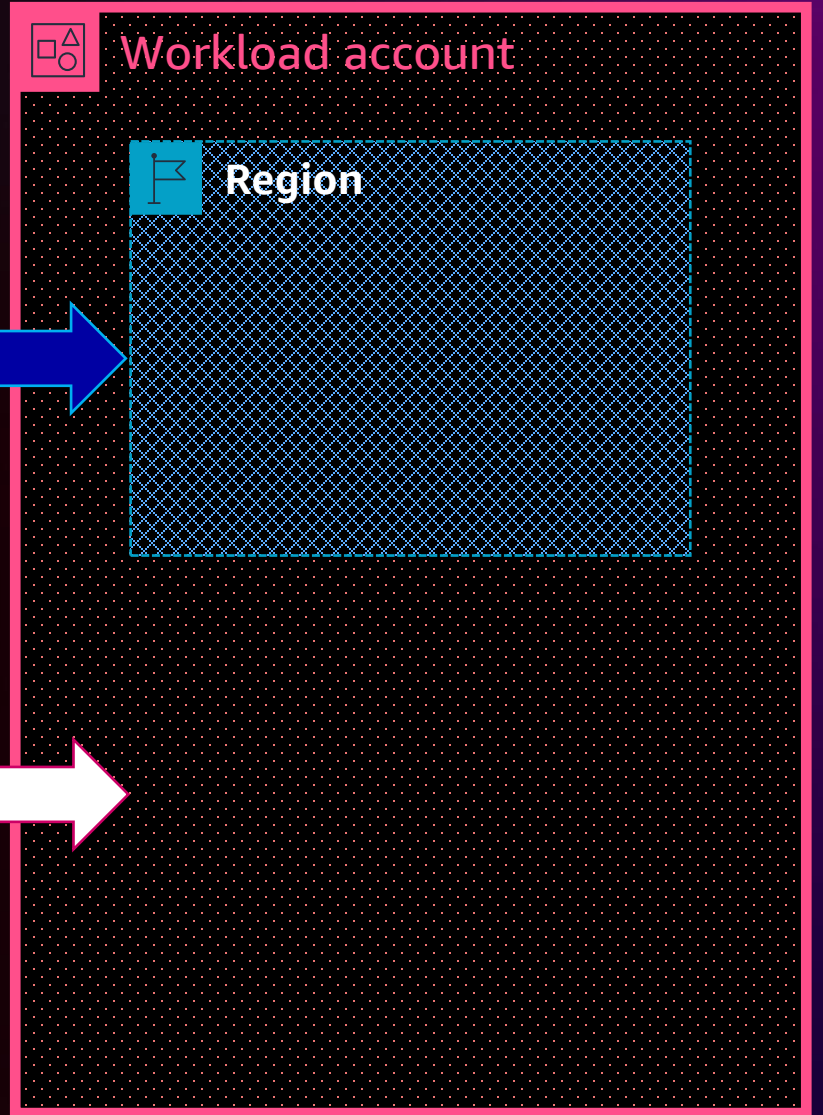
```
provider "aws" {  
  region = var.region  
  assume_role {  
    role_arn = "arn:aws:iam::${var.account}:role/BuildRole"  
    ...  
  }  
}
```

 dev eu-central-1/research/1.1

Deployment scope

```
module "regional" {  
  source = "../modules/regional"  
  ...  
}
```

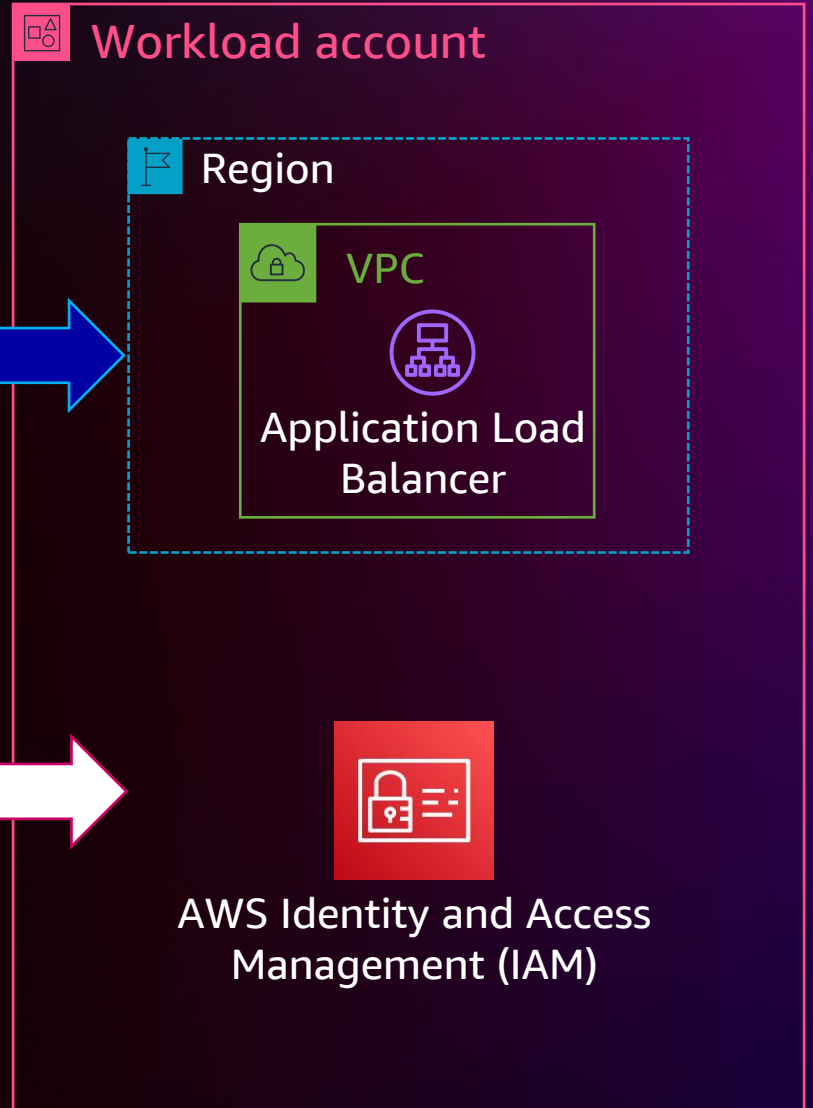
```
module "global" {  
  source = "../modules/global"  
  ...  
}
```



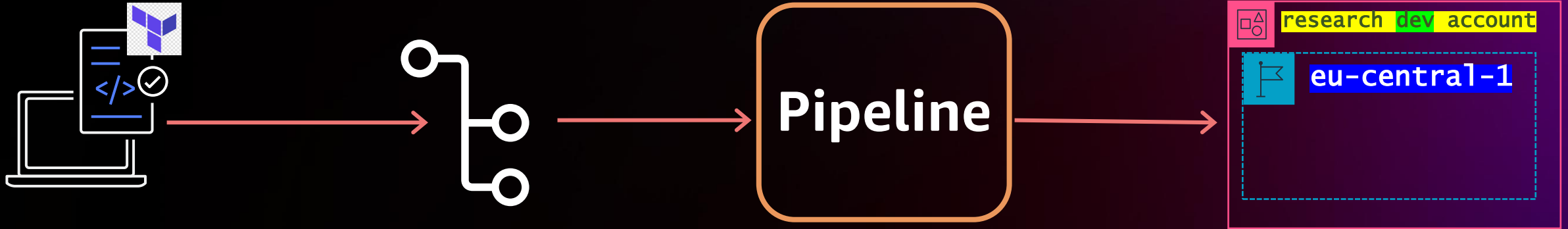
Sample

```
module "regional" {  
  source = "../modules/regional"  
  ...  
}
```

```
module "global" {  
  source = "../modules/global"  
  ...  
}
```



Deployment steps recap



`dev_eu-central-1/research`

1

Infra
blueprint

2

Git tag

3

Triggers
pipeline

4

Deploys into target
environment

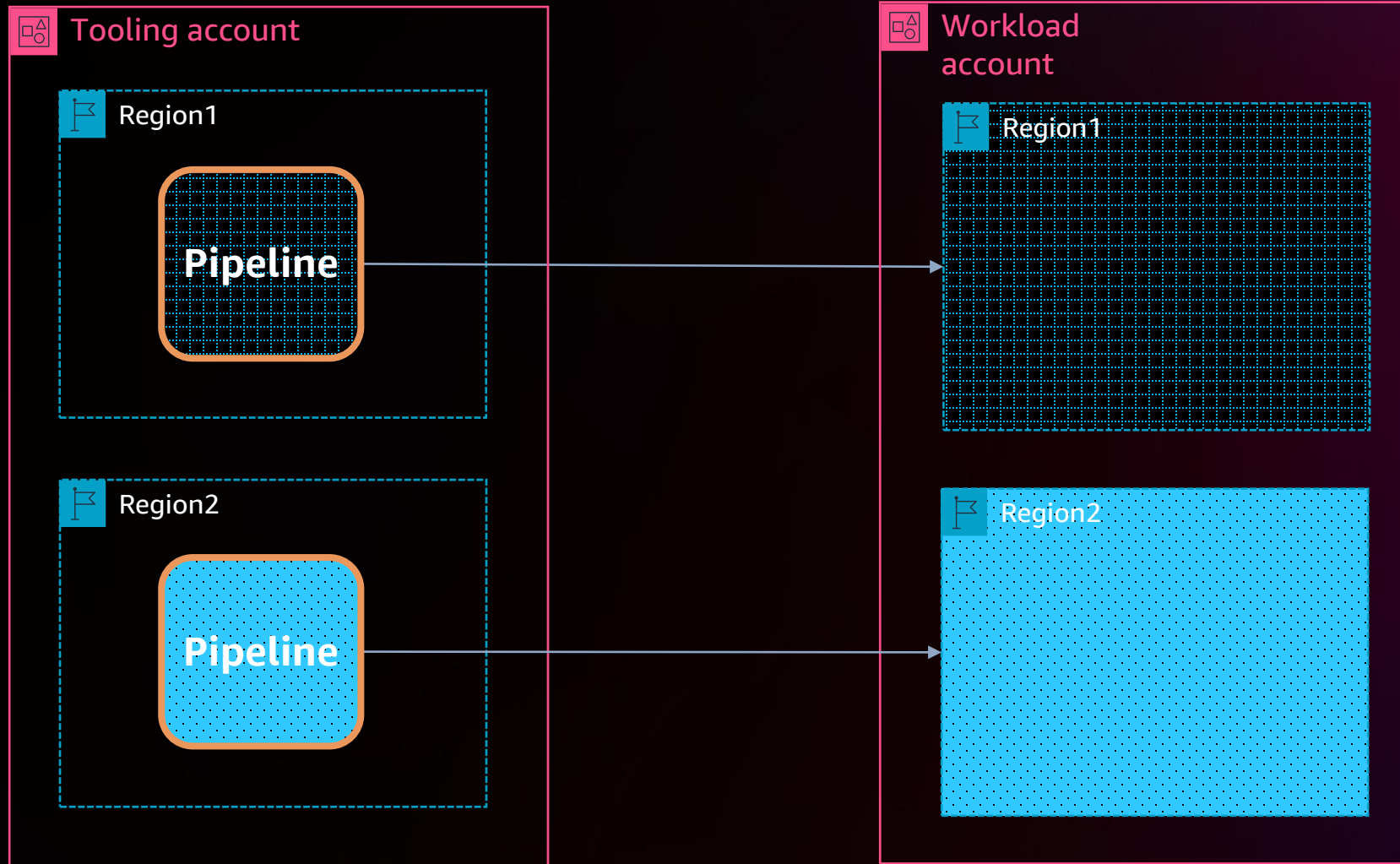
Pipeline considerations and best practices



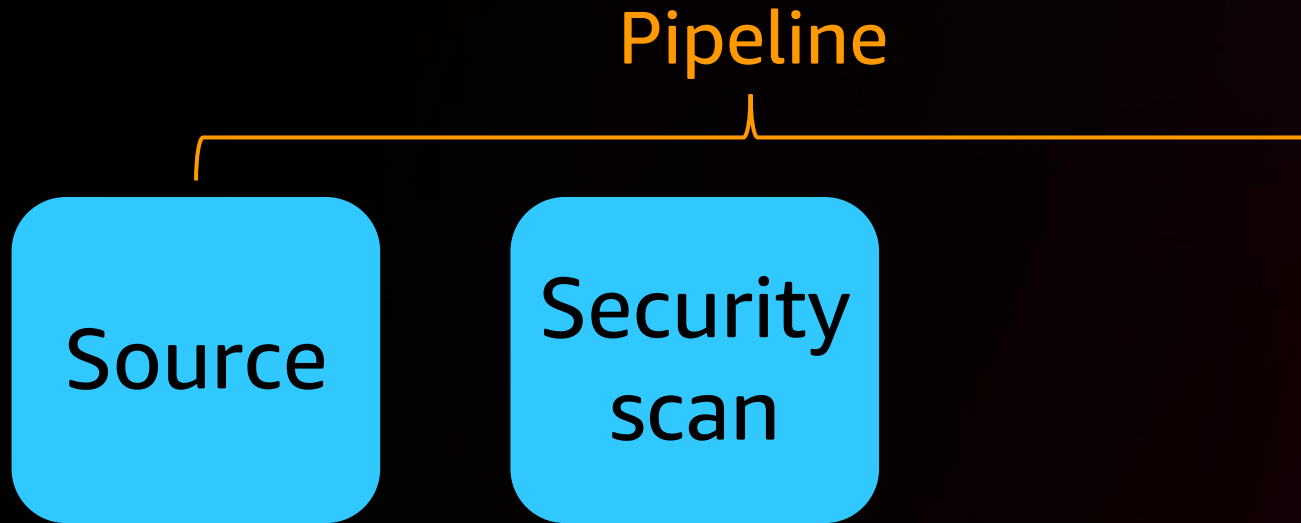
**Make the pipeline
the golden source
for deployments**



Multi-Region CI/CD pipeline



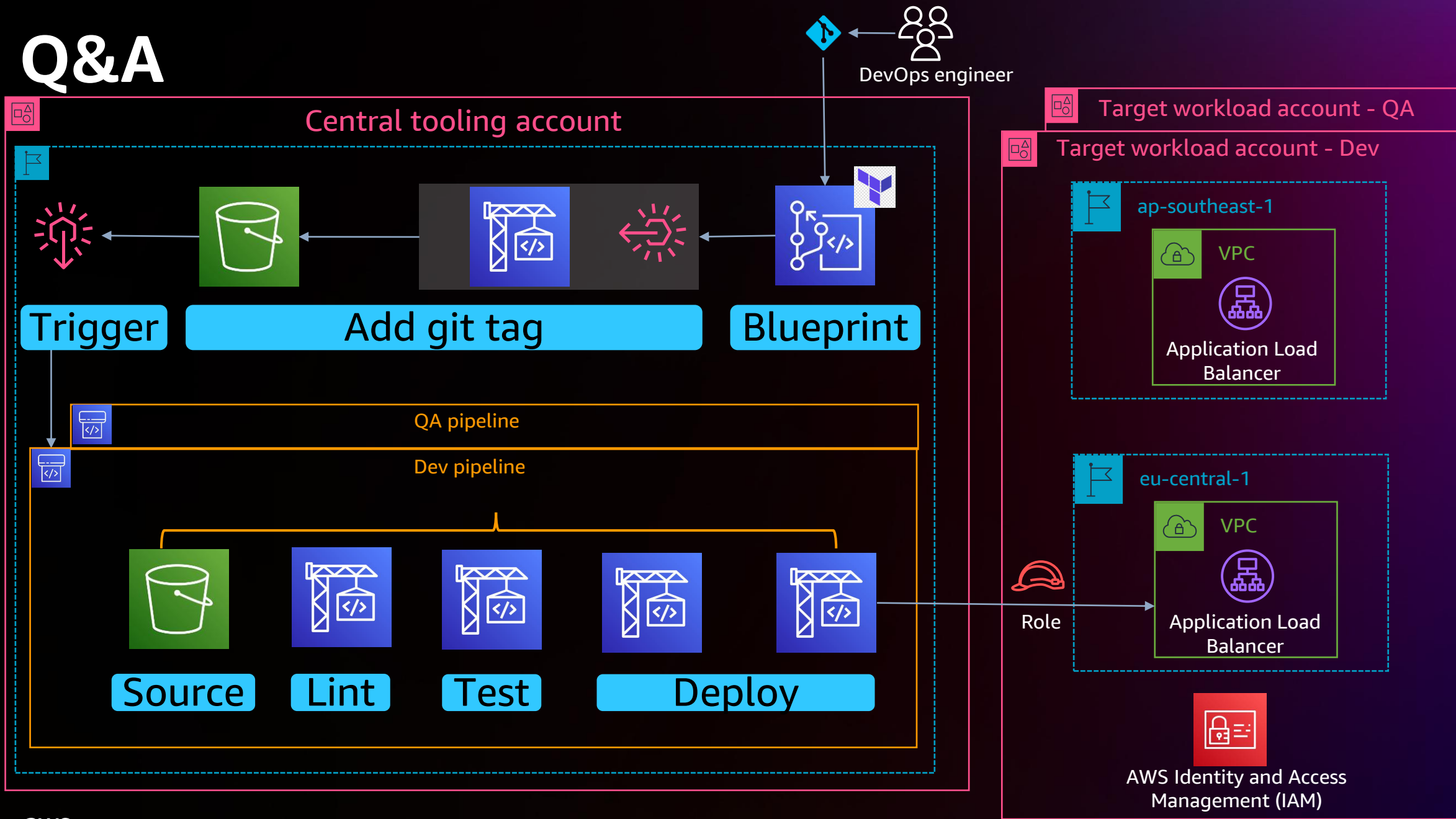
Shift security left in SDLC



Blog post and sample code



Q&A



Thank you!

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